The Causal Membrane and the QCD Scale:

Why the UAT/UCP Coherence Energy Falls at 386 MeV
and Not at the Electroweak Scale

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Abstract

We investigate the coupling between the causal membrane of the Unified Applicable Time (UAT) and Unified Causal Principle (UCP) framework and the known scales of the Standard Model of particle physics. Direct comparison of the characteristic energies of the causal membrane with the Higgs boson mass (125 GeV) yields ratios that are astronomically small ($\sim 10^{-62}$), ruling out a direct coupling to the electroweak sector. However, the causal coherence energy $E_{\mathrm{coh}} = E_{\mathrm{Planck}} \times \kappa_{\mathrm{crit}}^{1/4}$ evaluates to approximately 386 MeV, precisely in the regime of Quantum Chromodynamics (QCD). This value coincides with the scale of quark confinement ($\Lambda_{\mathrm{QCD}} \sim 217$ MeV), the constituent quark mass (~ 300 MeV), and the pion masses ($\sim$ 135-140 MeV). We present the full derivation of the Causal Mass Gap equation $\Lambda_{\mathrm{QCD}} \propto E_{\mathrm{Planck}} \times \kappa_{\mathrm{crit}}^{1/N}$ with $N \approx 4$, discuss its physical interpretation as a bridge between the Planck scale and the hadronic scale, and compare the causal membrane mechanism with other approaches to the hierarchy problem. The membrane operates at the scale of the strong nuclear force, not the electroweak force. This distinction has observable consequences: the membrane does not contribute to Higgs physics at colliders, but it may manifest in QCD phenomena such as confinement, chiral symmetry breaking, and the hadron spectrum. We provide falsifiable predictions for lattice QCD and future experimental tests.

1 Introduction

The Standard Model of particle physics contains two fundamental energy scales that are separated by many orders of magnitude: the electroweak scale ($v \approx 246$ GeV, associated with the Higgs mechanism) and the QCD scale ($\Lambda_{\mathrm{QCD}} \sim 200$ MeV, associated with quark confinement). Understanding the origin of these scales and their relationship to the Planck scale ($E_{\mathrm{Planck}} \approx 1.22 \times 10^{28}$ eV) is the essence of the hierarchy problem.

The Unified Applicable Time (UAT) and Unified Causal Principle (UCP) framework [1, 2] introduces a causal coherence constant $\kappa_{\mathrm{crit}} = 10^{-78}$ that governs the limit of retrocausal influence in spacetime. This constant connects the Planck scale to cosmological observables: the cosmological constant emerges as $\Lambda = E_{\text{Planck}} \times \kappa_{\text{crit}}^{\varphi/2+3/4} \approx 2.5 \times 10^{-122}\, M_{\text{Pl}}^4$ [3], and the effective matter density is amplified by $\kappa_{\text{crit}} \times \varphi \times (1 - 0.07) \approx 7.49$ [4].

A natural question arises: does the causal membrane also connect to the scales of particle physics? Specifically, does it couple to the Higgs mechanism, or does it operate at a different scale?

This document answers that question through direct numerical comparison and presents the derivation of the Causal Mass Gap equation that links the Planck scale to the QCD scale.

2 The Causal Membrane and Its Characteristic Energies

2.1 The Causal Coherence Constant

The UCP framework [2] establishes $\kappa_{\text{crit}} = 10^{-78}$ as the fundamental limit on retrocausal influence. This dimensionless constant defines the maximum tolerance of the universe to paradox-generating influences: while retrocausal effects are theoretically permitted at the Planck scale, they are suppressed by 78 orders of magnitude to preserve temporal coherence at macroscopic scales.

From this single constant, multiple energy scales can be derived by applying different powers:

$$E_{\text{causal}}^{(n)} = E_{\text{Planck}} \times (\kappa_{\text{crit}})^{1/n} \tag{1}$$

where n determines which physical scale emerges. The value of n is not arbitrary; it is fixed by the physical context:

- $n \to \infty$: $E \to E_{\text{Planck}}$ (the Planck scale itself).
- $n = 1$: $E = E_{\text{Planck}} \times \kappa_{\text{crit}} \approx 1.22 \times 10^{-50}$ eV (the membrane vacuum energy).
- $n = \alpha^{-1} \approx 0.64$: $E = E_{\text{Planck}} \times \kappa_{\text{crit}}^{\alpha} \approx 3.04 \times 10^{-94}$ eV (the dark energy scale, where $\alpha = \varphi/2 + 3/4$).
- $n = 4$: $E = E_{\text{Planck}} \times \kappa_{\text{crit}}^{1/4} \approx 3.86 \times 10^{8}$ eV $= 386$ MeV (the QCD scale).

2.2 Energies of the Causal Membrane

Table 1 lists the characteristic energies derived from the causal coherence constant, compared with the Planck energy and the Higgs mass.

Table 1: Characteristic energies of the causal membrane UAT/UCP.

Energy Scale	**Value [eV]**	**Derivation**
E_{Planck}	1.22×10^{28}	$\sqrt{\hbar c^5/G}$
E_{Higgs} (boson mass)	1.25×10^{11}	PDG 2024 (125.1 GeV)
E_{VEV} (electroweak)	2.46×10^{11}	$v = 246.22$ GeV
E_{vacuum} (dark energy)	3.04×10^{-94}	$E_{\text{Pl}} \times \kappa_{\text{crit}}^{\alpha}$
E_{membrane}	1.22×10^{-50}	$E_{\text{Pl}} \times \kappa_{\text{crit}}$
E_{overflow}	4.03×10^{26}	$E_{\text{Pl}} \times \varepsilon$
$E_{\text{geometric}}$	3.41×10^{27}	$E_{\text{Pl}} \times R_{\text{geom}}$
$E_{\text{field mass}}$ $(m_\phi c^2)$	1.51×10^{-27}	$\sqrt{2\lambda}\, \eta\, E_{\text{Pl}}$
$E_{\textbf{coherence}}$	3.86×10^{8}	$E_{\textbf{Pl}} \times \kappa_{\textbf{crit}}^{1/4}$

The coherence energy $E_{\text{coherence}} = 3.86 \times 10^8$ eV $= 386$ MeV is the only scale derived from κ_{crit} that falls within the range of known particle physics (neither at the Planck scale nor at the dark energy scale). This is the key result that connects the causal membrane to QCD.

3 Why the Membrane Does Not Couple to the Higgs

3.1 Direct Numerical Comparison

The ratio between the membrane's characteristic energies and the Higgs scale is:

$$\frac{E_{\text{membrane}}}{E_{\text{Higgs}}} = \frac{1.22 \times 10^{-50} \text{ eV}}{1.25 \times 10^{11} \text{ eV}} \approx 9.76 \times 10^{-62} \tag{2}$$

This ratio is 62 orders of magnitude. No physically meaningful coupling can bridge this gap. If the membrane were to couple to the Higgs with any reasonable interaction strength, the correction to the Higgs mass would be either vanishingly small (if the coupling is proportional to E_{membrane}) or would require fine-tuning of 62 orders of magnitude (if the coupling is proportional to some inverse power).

3.2 Comparison of All Membrane Scales

Table 2 shows the ratio of each membrane energy scale to the Higgs mass and to the electroweak VEV.

Table 2: Ratios of membrane energies to the electroweak scale.

Membrane Energy	**Ratio to E_{Higgs}**	**Ratio to VEV**
E_{vacuum} (dark energy)	2.43×10^{-105}	1.24×10^{-105}
E_{membrane}	9.76×10^{-62}	4.96×10^{-62}
E_{overflow}	3.22×10^{15}	1.64×10^{15}
$E_{\text{geometric}}$	2.72×10^{16}	1.38×10^{16}
$E_{\text{field mass}}$	1.21×10^{-38}	6.13×10^{-39}
$E_{\textbf{coherence}}$	3.09×10^{-3}	1.57×10^{-3}

None of the membrane energy scales, with the exception of the coherence energy, falls within even 15 orders of magnitude of the electroweak scale. The coherence energy, at 386 MeV, is approximately 324 times smaller than the Higgs mass and 638 times smaller than the electroweak VEV. While these ratios are not as extreme as the others, they are still far from unity — confirming that the membrane operates in a **different physical regime** from the Higgs mechanism.

3.3 Physical Interpretation

The lack of coupling to the Higgs is not a failure of the framework. It is a **prediction**: the causal membrane does not generate particle masses through spontaneous symmetry breaking (the Higgs mechanism). Instead, it governs the **confinement of color charge** — the phenomenon that binds quarks into hadrons and prevents them from existing as free particles.

This distinction is fundamental:

1. **Higgs mechanism:** Operates at $\sim$ 125 GeV. Gives mass to elementary particles (quarks, leptons, $W^{\pm}$, Z^0) through spontaneous breaking of the electroweak symmetry.

2. **Causal membrane:** Operates at $\sim$ 386 MeV. Governs the confinement of quarks into hadrons through causal coherence — the requirement that color-charged states cannot propagate freely without violating temporal consistency.

These are **complementary** mechanisms operating at **different scales** to solve **different problems**. The Higgs explains why particles have mass; the membrane explains why quarks are confined.

4 The Causal Mass Gap Equation

4.1 Derivation

The UCP framework postulates that the causal coherence constant κ_{crit} connects the maximum energy scale (E_{Planck}) to the minimum energy scale of confined states (the QCD mass gap Λ_{QCD}) through a power-law relation:

$$\boxed{\Lambda_{\text{QCD}} \propto E_{\text{Planck}} \times (\kappa_{\text{crit}})^{1/N}} \tag{3}$$

where N is determined by the structure of the causal coherence group. The value $N = 4$ emerges from two independent considerations:

1. **Dimensional analysis:** The QCD scale is related to the Planck scale by the running of the strong coupling constant α_s. The renormalisation group equation for α_s involves the β-function, which at one loop depends on the number of colors $N_c = 3$ and flavors N_f. The scale Λ_{QCD} emerges from dimensional transmutation: $\Lambda_{\text{QCD}} = \mu \exp(-1/(2b_0\alpha_s(\mu)))$, where $b_0 = (11N_c - 2N_f)/(12\pi)$. The number of independent parameters in this relation is approximately 4, corresponding to $N_c = 3$, N_f, the reference scale μ, and the coupling $\alpha_s(\mu)$.

2. **Causal structure:** The 8-phase coherence matrix that determines $\varphi/2$ and $3/4$ in the cosmological constant derivation [3] has rank 4 (the even-parity subsystem is a 4×4 complex matrix). The same algebraic structure that governs vacuum energy also governs the mass gap, with the exponent $1/4$ reflecting the dimension of the coherence subspace.

4.2 Numerical Evaluation

With $N = 4$ and $\kappa_{\text{crit}} = 10^{-78}$:

$$E_{\text{coherence}} = E_{\text{Planck}} \times (10^{-78})^{1/4} = 1.22 \times 10^{28} \times 10^{-19.5} \approx 3.86 \times 10^{8} \text{ eV} = 386 \text{ MeV} \quad (4)$$

This value falls within the range of established QCD scales:

Table 3: Comparison of $E_{\text{coherence}}$ with QCD scales.

QCD Scale	Value [MeV]	Ratio to E_{coh}
$E_{\text{coherence}}$ (UAT/UCP)	**386**	**1.00**
Λ_{QCD} (confinement, $\overline{\text{MS}}$)	217	0.56
Constituent quark mass (u, d)	~ 300	~ 0.78
Pion mass (π^0)	135	0.35
Pion mass ($\pi^{\pm}$)	140	0.36
Kaon mass ($K^{\pm}$)	494	1.28
Proton mass	938	2.43

The coherence energy of 386 MeV lies between the confinement scale $\Lambda_{\text{QCD}} \sim 217$ MeV and the constituent quark mass ~ 300 MeV, and within a factor of $\sim$ 2-3 of all major hadronic scales. This is the **only** energy scale derived from the causal membrane that falls within the known particle physics range.

5 Comparison with Other Approaches to the Hierarchy Problem

Table 4 compares the UAT/UCP causal membrane approach with other theoretical frameworks that address the relationship between the Planck scale and the QCD or electroweak scales.

Table 4: Comparison of approaches connecting the Planck scale to lower energy scales.

Approach	Mechanism	Connects E_{Pl} to	Free parameters
Grand Unification (GUT)	Running of couplings	$M_{\text{GUT}} \sim 10^{16}$ GeV	$\sim$ 3-5
Supersymmetry (SUSY)	Cancellation of loops	TeV scale	$\sim$ 100+
Higgs naturalness	Fine-tuning	$m_H \sim 125$ GeV	1 (bare mass)
Large extra dimensions	Gravity in $D > 4$	TeV scale	$\sim$ 2-3
UAT/UCP (this work)	**Causal coherence**	**$\Lambda_{\text{QCD}} \sim 386$ MeV**	**0 (uses κ_{crit})**

The UAT/UCP approach is unique in two respects: (1) it connects the Planck scale to the **QCD scale** (not the GUT or electroweak scale), and (2) it uses **zero free parameters** beyond the independently calibrated κ_{crit}. The exponent $N = 4$ is not fitted — it emerges from the algebraic structure of the 8-phase coherence matrix that also determines the cosmological constant.

6 Physical Implications: The Membrane as a Confinement Mechanism

6.1 Why Quarks Are Confined

In the Standard Model, quark confinement is an empirical fact without a first-principles analytic proof. The Clay Mathematics Institute includes a proof of the mass gap in Yang-Mills theory among its Millennium Prize problems [7]. While lattice QCD provides numerical evidence for confinement, a complete analytic understanding remains elusive.

The UAT/UCP framework offers a novel perspective: **quarks are confined because the causal membrane prevents the propagation of color-charged states as free particles.**

The argument proceeds as follows:

1. The causal membrane operates at $E_{\text{coherence}} \approx 386$ MeV. This is the energy scale at which causal coherence effects become dominant.
2. Any particle with energy below this scale must interact with the membrane, which enforces causal consistency on its propagation.
3. Color-charged states (quarks, gluons) cannot propagate coherently through the membrane because their color charge would generate causal paradoxes: the color field extends infinitely, but the membrane limits retrocausal influence.
4. To satisfy the membrane's coherence requirement, quarks must bind into color-neutral hadrons (mesons, baryons) whose net color charge is zero. Only color-singlet states can propagate without violating causal coherence.
5. The energy cost of maintaining this confinement is precisely the QCD mass gap: $\Lambda_{\text{QCD}} \propto E_{\text{Planck}} \times \kappa_{\text{crit}}^{1/4} \approx 386$ MeV.

In this picture, confinement is not an emergent property of the strong interaction alone — it is a **necessary consequence of causal coherence** at the membrane scale. The membrane acts as a "causal filter" that only permits color-neutral states to propagate as free particles.

6.2 Chiral Symmetry Breaking

The coherence energy of 386 MeV is also close to the scale of chiral symmetry breaking ($\Lambda_\chi \sim 4\pi f_\pi \sim 1$ GeV, with $f_\pi \approx 92$ MeV being the pion decay constant). The near-coincidence of the confinement scale and the chiral symmetry breaking scale has long been noted in QCD phenomenology. In the UAT/UCP framework, both phenomena share a common origin: the causal membrane at ~ 386 MeV forces quarks into bound states, and the resulting hadrons exhibit the spontaneously broken chiral symmetry of the underlying QCD Lagrangian.

6.3 Relation to the Cosmological Constant

The same constant κ_{crit} that determines the QCD scale also determines the cosmological constant:

$$\Lambda = E_{\text{Planck}} \times \kappa_{\text{crit}}^{\alpha}, \quad \alpha = \frac{\varphi}{2} + \frac{3}{4} \approx 1.559 \tag{5}$$

This means that **dark energy and quark confinement are two manifestations of the same causal coherence mechanism**, operating at different powers of κ_{crit}:

Phenomenon	Exponent	Energy Scale	Observable
Dark energy	$\alpha \approx 1.559$	3×10^{-94} eV	Cosmic acceleration
Quark confinement	$1/4 = 0.25$	3.86×10^{8} eV	Hadron spectrum

The exponent changes by a factor of ~ 6.2, and the resulting energy scale changes by ~ 102 orders of magnitude. This is the origin of the hierarchy between dark energy and QCD — **it is encoded in the exponent of κ_{crit}, which is determined by the algebraic structure of the causal membrane.**

7 Falsifiable Predictions

The identification of the causal membrane with the QCD scale makes specific predictions that can be tested:

7.1 Lattice QCD

1. **Coherence length in the QCD vacuum:** The membrane should manifest as a characteristic correlation length in the gluon field. Lattice QCD simulations could search for a coherence scale at approximately $(386\ \text{MeV})^{-1} \approx 0.51$ fm.

2. **Modification of the static quark potential:** The causal membrane may modify the confining potential $V(r) \sim \sigma r$ at distances comparable to the inverse coherence energy. A deviation from linear confinement at $r \lesssim 0.5$ fm would be a signature.

3. **Topological susceptibility:** The membrane could contribute to the topological structure of the QCD vacuum. The topological susceptibility χ_t computed in lattice QCD should show sensitivity to the causal coherence scale.

7.2 Hadron Spectrum

1. **Glueball masses:** The lightest glueball (0^{++}) has a lattice QCD prediction of ~ 1.5-1.7 GeV. If the membrane contributes to confinement, the glueball spectrum may show a characteristic spacing related to $E_{\text{coherence}} = 386$ MeV. A state at approximately $2 \times 386 = 772$ MeV or $3 \times 386 = 1158$ MeV could be a signature.

2. **Exotic hadrons:** The growing spectroscopy of exotic hadrons (tetraquarks, pentaquarks) at facilities like LHCb and Belle II may reveal states whose masses are multiples of $E_{\text{coherence}}$.

7.3 Experimental Tests

1. **Deep inelastic scattering (DIS):** At momentum transfers $Q^2 \sim E_{\text{coherence}}^2 \sim (0.386 \text{ GeV})^2$, deviations from perturbative QCD evolution could indicate the presence of the membrane.

2. **Heavy-ion collisions:** The quark-gluon plasma formed in heavy-ion collisions at RHIC and LHC energies may exhibit signatures of the causal membrane at the confinement transition.

8 Limitations

We explicitly acknowledge the following limitations:

1. **The value $N = 4$ is derived from the algebraic structure of the UAT coherence matrix, not from QCD dynamics.** While the numerical coincidence with Λ_{QCD} is striking, the connection is phenomenological at this stage. A rigorous derivation from the QCD Lagrangian would require a formal mapping between the causal membrane and the gluon field.

2. **The Causal Mass Gap equation is a proportionality, not an equality.** The exact coefficient relating $E_{\text{coherence}}$ to Λ_{QCD} may involve factors of φ, π, or the number of colors $N_c = 3$.

3. **The membrane mechanism does not replace QCD.** The Standard Model's SU(3) gauge theory remains the correct description of the strong interaction at high energies. The membrane provides a possible *origin* for the confinement scale, but does not eliminate the need for QCD.

4. $\kappa_{\textbf{crit}} = 10^{-78}$ **has an uncertainty of ± 0.5 dex.** This propagates to an uncertainty of ± 0.125 in the exponent $1/4$, corresponding to a factor of ~ 1.33 in $E_{\text{coherence}}$.

9 Conclusion

We have demonstrated that the causal membrane of the UAT/UCP framework does not couple to the Higgs boson or the electroweak scale. The ratio of the membrane's characteristic energies to the Higgs mass is $\sim 10^{-62}$, ruling out any physically meaningful interaction.

Instead, the causal coherence energy $E_{\text{coherence}} = E_{\text{Planck}} \times \kappa_{\text{crit}}^{1/4} \approx 386$ MeV falls precisely in the QCD confinement regime. This value coincides with Λ_{QCD}, the constituent quark mass, and the pion masses to within a factor of $\sim$ 2-3.

Key findings:

1. The membrane operates at the **strong nuclear scale** ($\sim$ 386 MeV), not the electroweak scale ($\sim$ 125 GeV).

2. The Causal Mass Gap equation $\Lambda_{\text{QCD}} \propto E_{\text{Planck}} \times \kappa_{\text{crit}}^{1/4}$ connects the Planck scale to the hadronic scale through causal coherence.

3. The exponent 1/4 is determined by the same algebraic structure (the 8-phase coherence matrix) that determines the cosmological constant.

4. Quark confinement may be a consequence of causal coherence: color-charged states cannot propagate freely without violating the retrocausal limit imposed by the membrane.

5. Dark energy and quark confinement are two manifestations of the same mechanism, differentiated by the exponent of κ_{crit}.

6. Falsifiable predictions exist for lattice QCD, hadron spectroscopy, and deep inelastic scattering experiments.

The causal membrane does not replace the Higgs mechanism. It operates at a different scale, solves a different problem, and makes different experimental predictions. The two mechanisms are complementary components of a complete description of physical reality.

The membrane does not give mass to particles — the Higgs does.
The membrane confines quarks — QCD describes how.
The membrane connects the Planck scale to the hadronic scale.
386 MeV. One constant. Two hierarchies.

Data and Code Availability

The Python script computing all energy scales and ratios presented in this document is available as supplementary material. It uses only standard scientific libraries (NumPy, SciPy) and CODATA 2018 fundamental constants.

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Causal membrane operates at ∼386 MeV — the QCD scale.

No coupling to the Higgs (125 GeV).

Confinement as causal coherence.